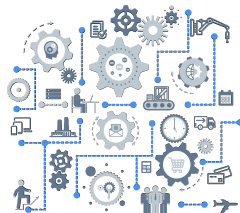
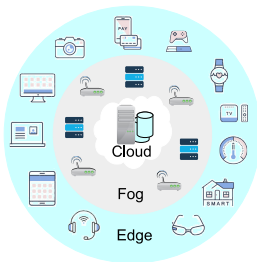


Privacy and Data Protection in Emerging Scenarios

Security, Privacy, and Data Protection Laboratory
Dipartimento di Informatica
Università degli Studi di Milano

ICT ecosystem

- Advancements in the **ICT** and **networks** have **changed** our society
- **5G and beyond**, **infrastructures** and **services** are more **powerful**, **efficient**, and **complex**

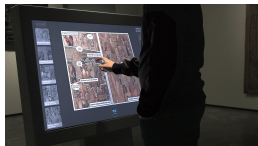


- **ICT** and **network** advancements are **enabling factors** for a **smart society** ...

... Everything is getting smart



Smart car



Museum and exhibitions



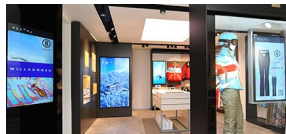
Health Care



Augmented reality



Smart e-commerce



Intelligent shops



Smart entertainment systems



Smart governance



Smart toothbrush

Smart society



Smart society - Advantages



Smart services and security – Advantages

- + Better protection mechanisms
- + Business continuity and disaster recovery
- + Prevention and response

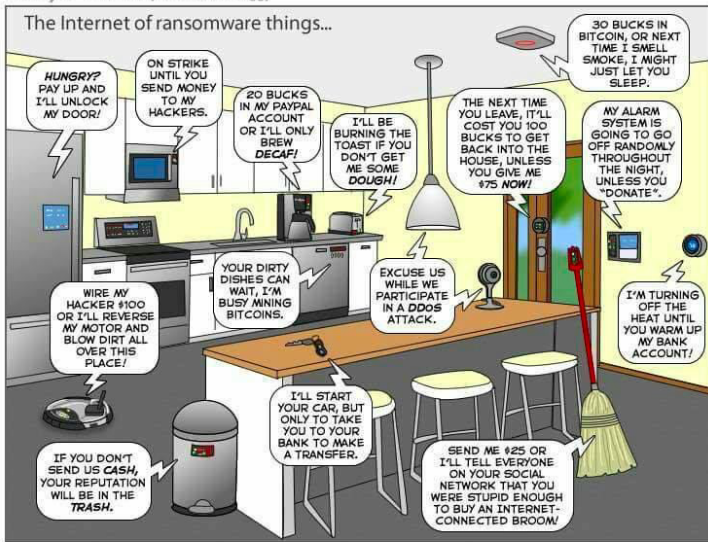
... but ...

Smart services and security – Disadvantages

- More complexity ...
 - ... weakest link becomes a point of attack
 - system hacking
 - improper information leakage
 - data and process tampering
- Explosion of damages and violations
- Loss of control over data and processes

Maybe too smart? – 1

The Joy of Tech™ by Nitrozac & Snaggy



You can help us keep the comics coming by becoming a patron!
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joyoftech.com

Maybe too smart? – 2



Security ... a complex problem



Protection of infrastructure



Protection of communication/network



Protection of devices

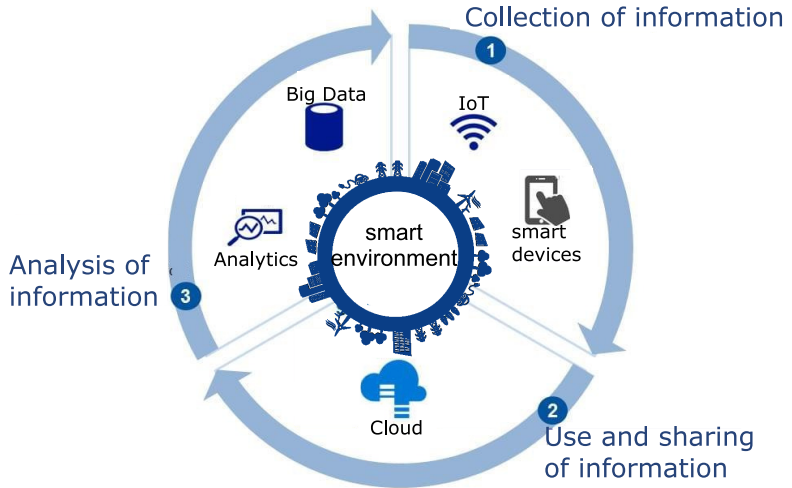


Protection against malware and attacks



Protection of data

The role of data in a smart environment

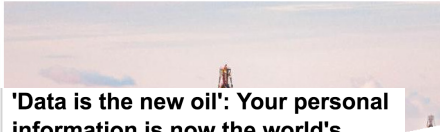


⇒ better governance and intelligent systems

The most valuable resource - Data

INQUIRER

The new oil: data is the world's most valuable resource



'Data is the new oil': Your personal information is now the world's most valuable commodity

Huge amounts of data are controlled by just 5 global mega-corporations that bigger than most governments

By Ramona Pringle, CBC News | Posted: Aug 25, 2017 5:00 AM ET | Last Updated: Aug 25, 2017 11:28 AM

Big Data and Analytics Play an Important Role in the Energy Industry

RealTimeDAILY

AROUND THE NET

Data Is Now The World's Most Valuable Resource

The Economist, Monday, May 8, 2017 6:22 AM

Data is now the world's most valuable resource according to *The Economist*, which reports on antitrust concerns about Alphabet (Google's parent company), Amazon, Apple, Facebook, and Microsoft, all of which have tons of data. The

Fuel of the future

Data is giving rise to a new economy

How is it shaping up?



Why is data protection so important?

06 February 2017

digitally needs to be properly protected. From financial information for your staff, data usage in the UK is protected by legal necessity, but crucial to protecting and maintaining your

PARTNER CONTENT ARVIND SINGH

IS BIG DATA THE NEW BLACK GOLD?

Impact on data protection and privacy

Uber reveals 2.7 million UK users of its app were affected by a mass data breach that saw names, emails and phone numbers stolen

- Uber has revealed 2.7m UK users of its app were affected by a 2016 data breach
- The taxi-hailing firm then tried to cover up the breach for more than a year
- It was also found Uber had paid two hackers £75,000 to delete the data

By The

PUBLI

Over 100GB of Secret Consumer Credit Data Leaked Online

A collection of 1.4 Billion Plain-Text leaked credentials is available online

December 12, 2017 By Pierluigi Paganini

f My Page

Like 2

G+

A 41-gigabyte archive containing **1.4 Billion** credentials in clear text was found in dark web, it had been updated at the end of November

Computer Scientists Develop a Simple Tool to Tell If Websites Suffered a Data Breach

Published: December 12, 2017.

Uber says data breach compromised 380K users in Singapore

Ride-sharing company reveals 380,000 in Singapore were affected by the massive data breach that compromised 57 million accounts globally, but says no fraud or misuse has been tied to these users.

By Cyber Via The Way | December 08, 2017 -- 03:07 GMT-08:00 (GMT) | Tags: Security



NEWS

63,500 records breached by misconfigured database

by Jessica Davis | April 12, 2018

California Voters Suffer Major Data Breach

Step 01 2018

Equifax discovers another 2.4 million customers hit by data breach

Posted by Dissent at 11:02 am

Business Sector, Hack, U.S.

Deloitte hit by cyber-attack revealing clients' secret emails

Exclusive: hackers may have accessed usernames, passwords and personal details of top accountancy firm's blue-chip clients



Privacy

Carphone Warehouse Breach: 'Striking' Failures Trigger Fine

Matthew J. Schwartz • January 10, 2018

Mobile phone retailer Carphone Warehouse has been hit with one of the largest fines ever imposed by Britain's data privacy watchdog

The Dutch Data Protection Authority accidentally leaked its employees' data



by MK -- 4 weeks

Approx. 9,000 Penn students affected by security breach that released their private information

By Kelly Holmsheid | 05/12/18 6:56pm

SECURITY

MASSIVE

Personal Data of Over 143 Million Americans Stolen from a Credit Reporting Firm



By R

Se

18

SHARES

f SHARE

Tweet

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SHARE

MyFitnessPal breach affects millions of Under Armour users

Photo: Jeremy Bernhard, KSDN-TV

Former nursing home employee admits stealing residents' credit card numbers

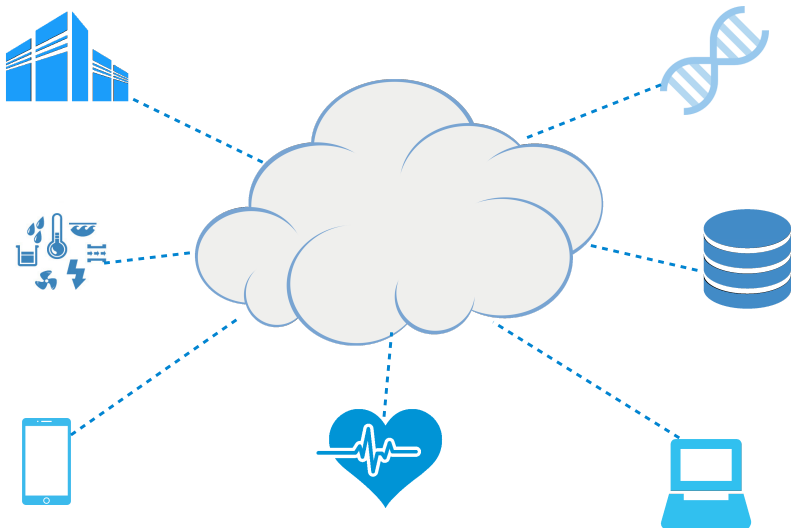
Shaniece Borney, 29, will be forced to pay the victims back and could face an additional \$250,000 fine, 10 years in prison or both.

NEWS

Facebook admits to far higher number of data breaches

Facebook has said personal data on 87 million users was shared with Cambridge Analytica, millions more than it admitted earlier. The social media giant also unveiled new privacy rules, but the whiff of scandal lingers.

Huge amount of data stored at external providers



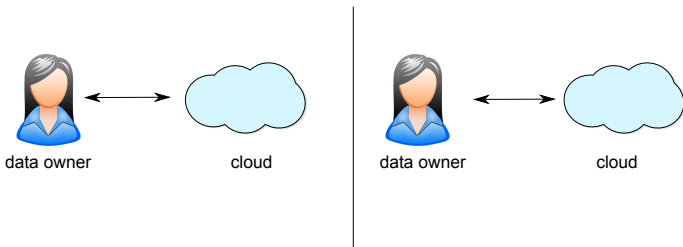
Cloud computing

- The Cloud allows users and organizations to rely on external providers for storing, processing, and accessing their data
 - + high configurability and economy of scale
 - + data and services are always available
 - + scalable infrastructure for applications
- Users lose control over their own data
 - new security and privacy problems
- Need solutions to protect data and to securely process them in the cloud



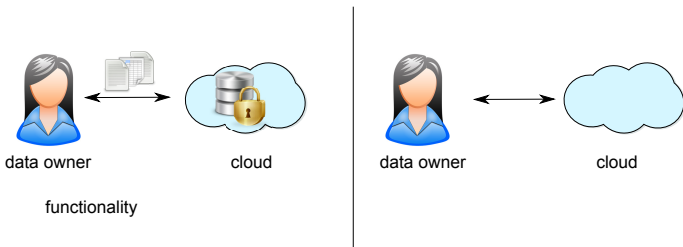
Cloud computing: Today

Cloud Service Providers (CSPs) apply security measures in the services they offer **but** these measures protect only the perimeter and storage against outsiders



Cloud computing: Today

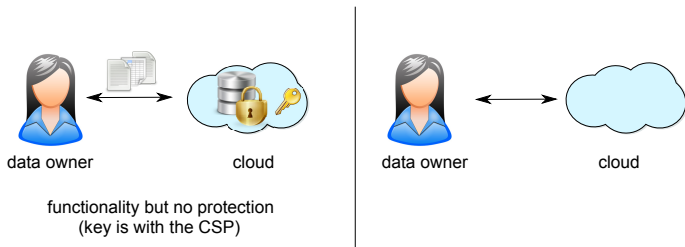
Cloud Service Providers (CSPs) apply security measures in the services they offer **but** these measures protect only the perimeter and storage against outsiders



- functionality

Cloud computing: Today

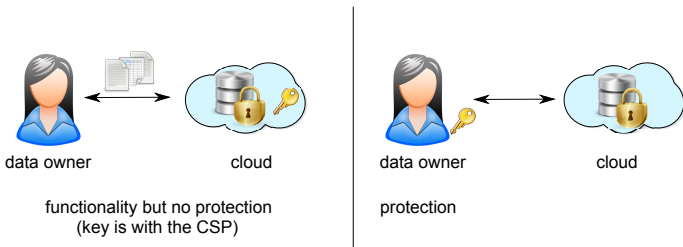
Cloud Service Providers (CSPs) apply security measures in the services they offer **but** these measures protect only the perimeter and storage against outsiders



- functionality implies **full trust in the CSP** that has full access to the data (e.g., Google Cloud Storage, iCloud)

Cloud computing: Today

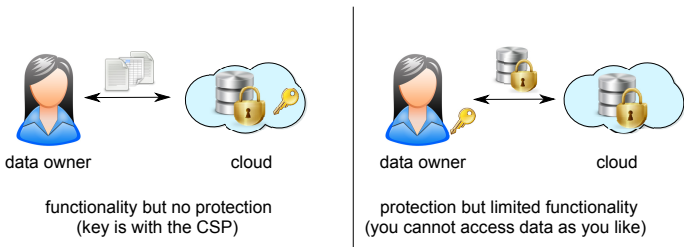
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- functionality implies **full trust in the CSP** that has full access to the data (e.g., Google Cloud Storage, iCloud)
- protection

Cloud computing: Today

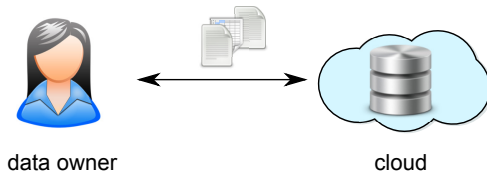
Cloud Service Providers (CSPs) apply security measures in the services they offer **but** these measures protect only the perimeter and storage against outsiders



- functionality implies **full trust in the CSP** that has full access to the data (e.g., Google Cloud Storage, iCloud)
- protection but **limited functionality** since the CSP cannot access data (e.g., Boxcryptor, SpiderOak)

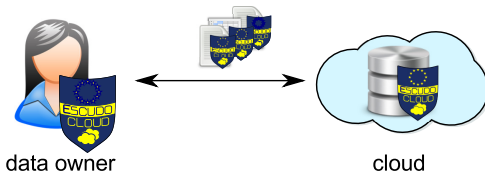
Cloud computing: New vision

Solutions that provide protection guarantees giving the data owners both: full control over their data and cloud functionality over them



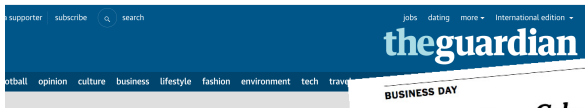
Cloud computing: New vision

Solutions that provide protection guarantees giving the data owners both: full control over their data and cloud functionality over them



- client-side trust boundary: only the behavior of the client should be considered trusted
⇒ techniques and implementations supporting direct processing of encrypted data in the cloud

Data protection – Base level



Yahoo hack: 1bn accounts compromised by biggest data breach in history

The latest incident to emerge - which happened in 2013 - is probably distinct from the breach of 500m user accounts in 2014

Technology

Hackers steal 2.5 million PlayStation and Xbox players' details in major breach



BUSINESS DAY

Equifax Says Cyberattack May Have Affected 143 Million in the U.S.

by CARA SIEGEL, BERNARD, TIFFANY HSU, NICOLE PERLROTH and RON LIEBER. SEPT. 7, 2017

the three major consumer credit bureaus on Thursday

Deloitte hit by cyber-attack revealing clients' secret emails

Exclusive: hackers may have accessed usernames, passwords and details of top accountancy firm's blue-chip clients

The Register
Biting the hand that feeds IT

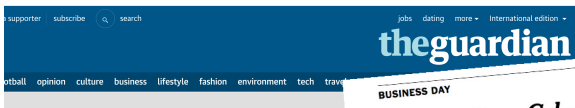


DATA CENTRE SOFTWARE SECURITY TRANSFORMATION DEVOPS BUSINESS PERSONAL TECH

Security

Two million customer records pillaged in IT souk CeX hack attack

Data protection – Base level



Yahoo hack: 1bn accounts compromised by biggest data breach in history

The latest incident to emerge - which happened in 2013 - is probably distinct from the breach of 500m user accounts in 2014

Technology

Hackers steal 2.5 million PlayStation and Xbox players'

Healthcare IT News

Privacy & Security

Even with encryption, EMR data at risk

'While encryption could offer some protections ... it also has serious limitations'

BUSINESS DAY

Equifax Says Cyberattack May Have Affected 143 Million in the U.S.

BY KARA SIEGEL, BERNARD, TIFFANY HSU, NICOLE PERLROTH and RON LIEBER SEPT. 7, 2017

Three major consumer credit bureaus on Thursday

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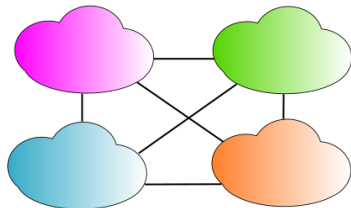
Security

Two million customer records pillaged in IT souk CeX hack attack

Data protection – Regulation



Access and usage control



Selective sharing



Governance and regulation

Data protection – Confidentiality (1)

- Minimize release/exposition
 - correlation among different data sources
 - indirect exposure of sensitive information
 - de-identification $\neq$ anonymization



TECHNOLOGY | UNBOXED

Big Data Is Opening Doors, but Maybe Too Many

By STEVE LOHR | MARCH 20, 2013

IN the 1960s, mainframe computers posed a significant technological challenge to common notions of privacy. That's when the federal government started putting tax returns into those giant machines, and consumer credit bureaus began building databases containing the personal financial information of millions of Americans. Many people feared that the new computerized databanks would be put in the service of an intrusive corporate or government Big Brother.

Data protection – Confidentiality (2)



The Telegraph

Home News World Sport Finance Comment Blogs Culture Travel Life Women
Technology News Technology Companies Technology Reviews Video Games Technology

HOME » TECHNOLOGY » FACEBOOK

Gay men 'can be identified by their Facebook friends'

Homosexual men can be identified just by looking at their Facebook friends, a unpublished research by two students at the Massachusetts Institute of Tec



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Privacy loophole found in genetic databases

DNA donors' identities can be determined from publicly available records.

Erika Check Hayden

17 January 2013

[Rights & Permissions](#)

A Face Is Exposed for AOL Searcher No. 4417749

By MICHAEL BARBARO and TOM ZELLER Jr.
Published: August 9, 2008

Buried in a list of 20 million Web search queries collected by AOL and recently released on the Internet is user No. 4417749. The number was assigned by the company to protect the searcher's anonymity, but it was not much of a shield.



No. 4417749 conducted hundreds of searches over a three-month period on topics ranging from "numb fingers" to "60 single men" to "dog that urinates on everything."

And search by search, click by click, the identity of AOL user No. 4417749 became easier to discern. There are queries for "landscapers in Lilburn, Ga," several people with the last name Arnold and "homes sold in shadow lake

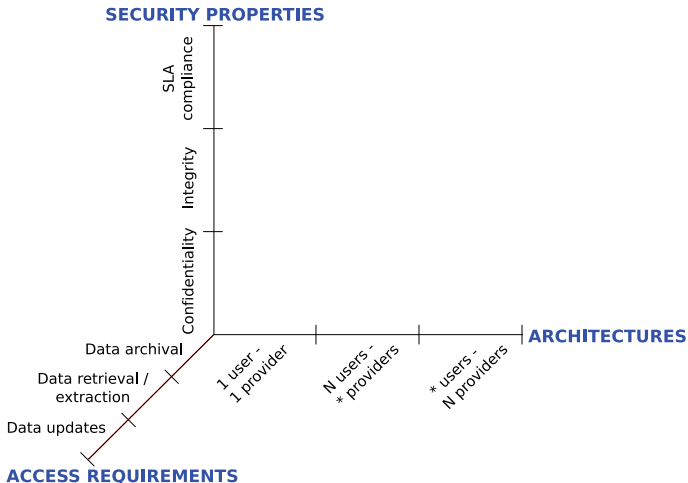
SIGN IN TO
E-MAIL THIS
PRINT
REPRINTS

THE
WAY WAY BACK
WATCH TRAILER

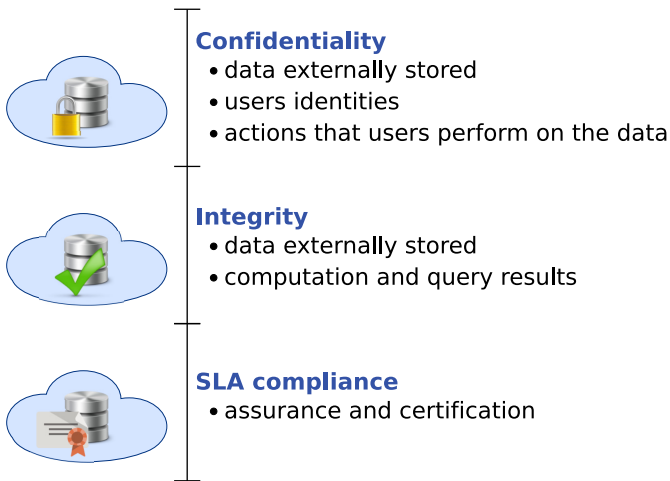
Characterization of Data Protection Challenges in Cloud Scenarios

Scientific and technical challenges

Three dimensions characterize the problems and challenges



Security properties



Access requirements



Data archival

- upload/download
- protection of data in storage



Data retrieval/extraction

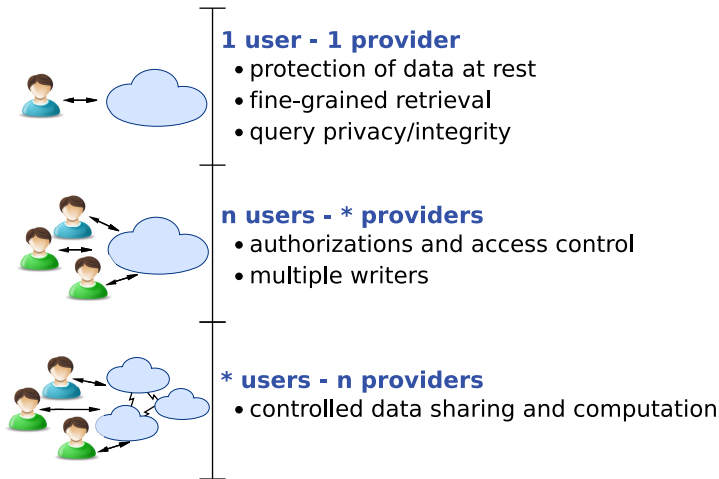
- support for fine-grained data retrieval and queries
- protection of computations and query results



Data update

- support for access retrieval and enforcement of updates
- protection of the actions and of their effects on the data

Architectures

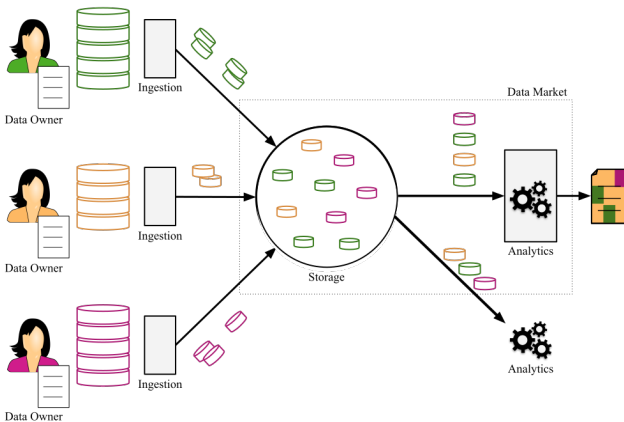


Combinations of the dimensions

- Every combination of the different instances of the dimensions identifies new problems and challenges
- The **security properties** to be guaranteed can depend on the **access requirements** and on the **trust assumption** on the providers involved in storage and/or processing of data
- Providers can be:
 - **curious**
 - **lazy**
 - **malicious**

Digital Data Market

Digital Data Market



Dimensions of the problem and challenges

- Requirements capturing and representation
policies regulating access, sharing, usage and processing

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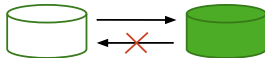
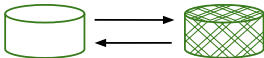
- Enforcing technologies
data wrapping / sanitization

Dimensions of the problem and challenges

- Requirements capturing and representation
policies regulating access, sharing, usage and processing



- Enforcing technologies
data wrapping / sanitization

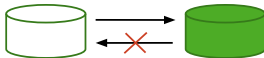
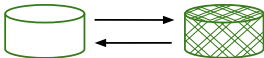


Dimensions of the problem and challenges

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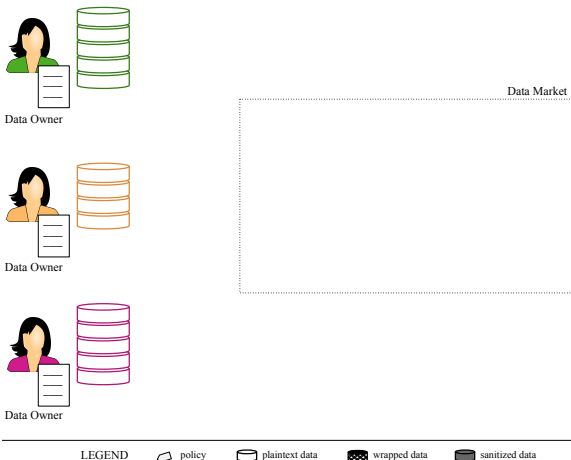
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- Enforcement phase
ingestion / storage / analytics

Enforcement phase

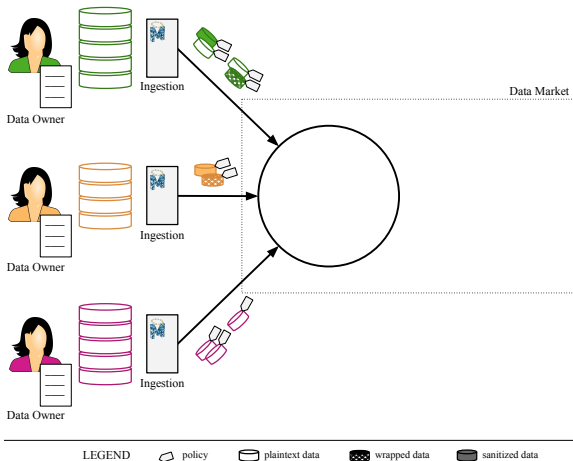
- Ingestion / Storage / Analytics



Multi-Owner data Sharing for Analytics and Integration respecting Confidentiality and OWNeR control - <https://mosaicrown.eu>

Enforcement phase

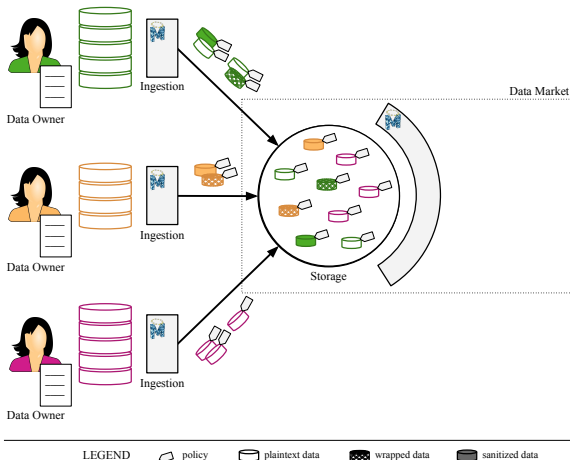
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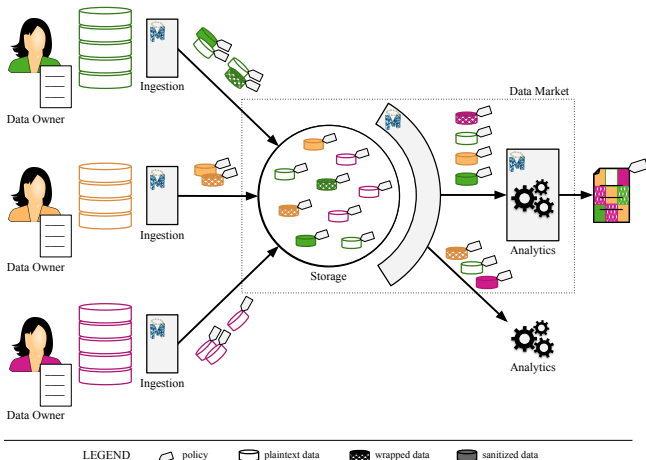
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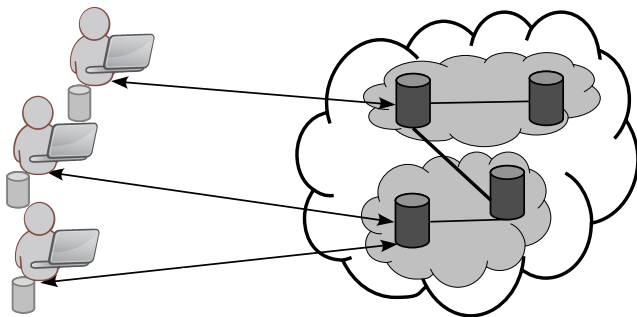
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Some Challenges in Data Protection

Issues to be addressed

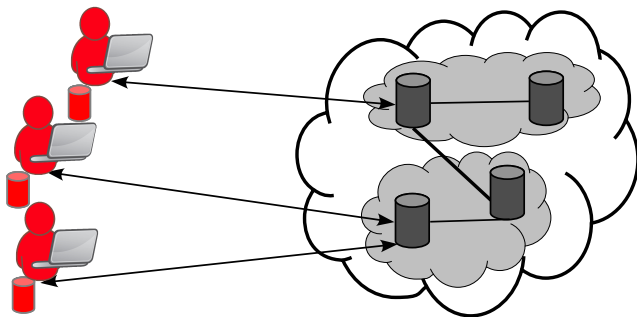
- Privacy of users
- Data protection
- Query execution
- Private access
- Data integrity and correctness
- Access control enforcement
- Data publication and utility
- Collaborative query execution

Security and privacy problems



S. De Capitani di Vimercati, S. Foresti, P. Samarati, "Protecting Data and Queries in Cloud-Based Scenarios," in *SN Computer Science*, vol. 4, n. 5, September 2023.

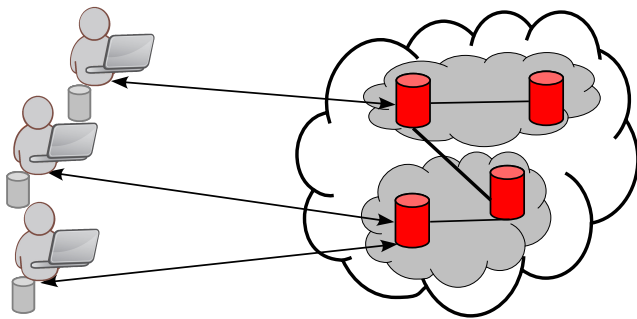
Security and privacy problems



Privacy of users

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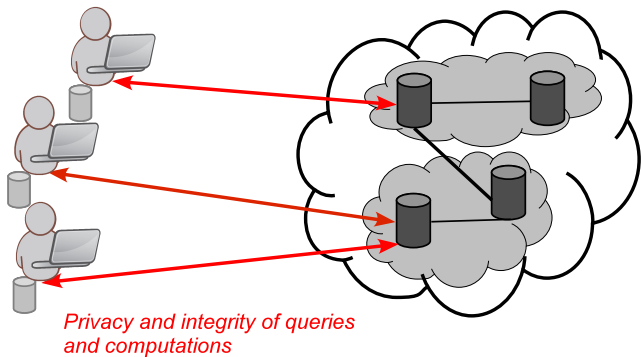


Privacy of users

Privacy and integrity of data storage

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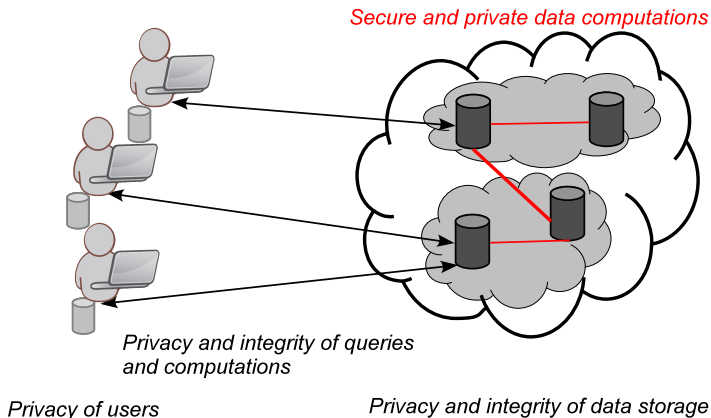


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